

KOKEMAKI TULKKILA, Finland

WMO: 029370

Lat: 61.2531N

Lon: 22.3461E

Elev: 128

StdP: 14.63

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.5	(c) -0.1	(d) -9.7	(e) 3.3	(f) -5.4	(g) -4.2	(h) 4.4	(i) 0.2	(j) 21.2	(k) 30.5	(l) 19.3	(m) 29.8	(n) 2.8	(o) 130	(p) 0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.2	(c) 80.7	(d) 64.6	(e) 76.8	(f) 63.2	(g) 73.4	(h) 61.5	(i) 67.3	(j) 76.3	(k) 65.2	(l) 73.9	(m) 63.5	(n) 71.3	(o) 7.8	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 64.4	(b) 90.9	(c) 70.0	(d) 61.8	(e) 82.9	(f) 68.3	(g) 60.1	(h) 78.0	(i) 66.8	(j) 31.7	(k) 76.5	(l) 30.2	(m) 74.3	(n) 28.9	(o) 71.6	(p) 73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	16.2	14.3	-7.0	83.9	8.5	5.2	-13.1	87.7	-18.1	90.7	-22.8	93.7	-29.0	97.5
			-7.4	68.7	8.3	2.8	-13.4	70.7	-18.3	72.3	-23.0	73.9	-29.1	75.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	21.2	25.8	29.3	39.7	51.1	57.0	62.1	59.9	52.0	40.8	35.5	28.8		
	DBStd	15.55	11.53	9.36	8.56	5.93	7.86	6.03	5.46	4.97	6.11	6.71	6.16	9.00		
	HDD50	4051	894	678	643	314	81	9	0	0	49	292	434	657		
	HDD65	8450	1359	1098	1108	758	436	249	125	171	390	750	884	1122		
	CDD50	1138	0	0	0	6	114	219	374	308	110	7	0	0		
	CDD65	62	0	0	0	0	4	9	34	14	1	0	0	0		
	CDH74	680	0	0	0	0	79	107	369	123	2	0	0	0		
	CDH80	133	0	0	0	0	7	15	86	25	0	0	0	0		
	WSAvg	6.9	6.7	7.1	7.4	6.8	7.1	7.3	6.4	6.6	6.4	6.9	7.1	7.5		
	PrecAvg	24.7	1.9	1.3	1.3	1.4	1.5	2.4	2.8	2.7	2.3	2.6	2.4	2.2		
(15) Precipitation	PrecMax	32.6	3.7	3.0	2.4	2.5	3.9	4.8	4.9	8.3	5.0	5.7	5.1	4.3		
	PrecMin	19.9	0.6	0.1	0.3	0.2	0.3	0.6	0.2	0.8	0.6	0.3	0.4	0.7		
	PrecStd	2.8	0.7	0.7	0.5	0.6	0.9	0.9	1.3	1.5	1.3	1.4	0.9	0.9		
	PrecStd	2.8	0.7	0.7	0.5	0.6	0.9	0.9	1.3	1.5	1.3	1.4	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	42.3	51.2	68.6	80.8	82.3	86.7	84.2	72.5	58.1	51.1	48.7		
		MCWB	38.5	37.8	42.2	52.2	60.7	66.1	67.7	68.9	61.5	53.0	49.5	47.0		
	2%	DB	38.3	39.7	46.4	61.2	75.6	76.8	82.1	76.9	68.4	54.6	47.7	43.1		
		MCWB	36.8	37.1	40.0	48.9	58.4	63.0	66.0	64.5	60.0	51.7	46.4	41.1		
	5%	DB	36.0	37.8	42.1	55.7	71.5	72.9	78.1	73.3	64.9	53.0	45.5	40.5		
		MCWB	34.8	36.2	37.5	45.5	56.7	60.7	64.8	62.3	57.7	50.7	44.5	38.9		
	10%	DB	34.3	36.1	39.2	51.2	66.9	69.7	73.8	70.1	62.2	51.1	43.7	38.5		
		MCWB	33.4	34.9	35.8	43.3	54.4	58.3	62.7	60.8	56.9	48.9	42.6	37.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.3	38.9	43.8	54.1	63.7	68.0	71.0	69.7	63.4	54.4	49.9	47.2		
		MCDB	40.1	40.6	48.9	63.7	74.7	76.9	81.5	78.4	70.0	56.8	50.8	48.7		
	2%	WB	37.2	37.5	40.4	50.0	60.8	64.7	68.1	65.9	60.8	52.8	46.7	41.7		
		MCDB	38.2	39.0	45.4	61.1	71.8	74.1	77.2	73.5	66.2	54.4	47.5	43.2		
	5%	WB	35.2	36.4	38.1	46.5	58.2	62.4	66.2	64.0	59.1	51.2	44.7	39.2		
		MCDB	35.9	37.9	41.4	54.2	69.2	71.1	75.5	71.1	63.4	52.7	45.5	40.4		
	10%	WB	33.6	35.1	36.4	44.1	55.4	60.2	64.4	62.2	57.5	49.3	42.9	37.5		
		MCDB	34.3	36.3	38.9	50.3	66.2	67.7	72.7	68.0	61.2	50.9	43.7	38.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	9.6	13.6	17.1	20.1	17.1	18.2	17.2	15.2	11.2	6.6	7.9		
		MCDBR	6.3	9.4	15.8	26.8	28.4	23.8	25.2	23.3	20.0	12.1	8.1	8.8		
		MCWBR	6.0	7.8	10.7	15.8	13.8	12.1	11.3	11.8	12.3	9.8	7.7	7.9		
		MCDBR	6.2	8.5	13.9	23.6	23.7	20.7	21.4	18.8	16.7	10.1	8.3	8.4		
	5% WB	MCWBR	6.1	7.2	9.9	14.6	12.3	11.5	10.6	10.4	10.9	9.2	7.8	7.8		
		MCWBR	6.1	7.2	9.9	14.6	12.3	11.5	10.6	10.4	10.9	9.2	7.8	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.306	0.329	0.368	0.364	0.356	0.384	0.370	0.344	0.327	0.285	0.359		
	taud		2.130	2.224	2.308	2.330	2.405	2.451	2.396	2.426	2.489	2.398	2.199	2.564		
	Ebn at Noon		143	210	250	261	272	277	264	259	246	204	133	95		
	Edh at Noon		12	21	27	33	33	32	33	30	23	18	11	8		
(44) All-Sky Solar Radiation	RadAvg		69	249	721	1191	1630	1741	1631	1278	767	326	86	40		
	RadStd		12	54	88	131	163	131	160	135	94	60	16	4		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	-243	N/A	N/A	N/A

Nomenclature: See separate page